Universal Health Care in Developed vs Developing Countries: A Broad Cross-Country Analysis

ECON 498 H

Professor Gabrielle Vasey

Srimayee Agnihotram

Student ID: 40243317

Department of Economics

Concordia University

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Introduction

The Universal Healthcare Coverage (UHC) policy is a tool aimed to achieve the 3rd Sustainable Development Goal (SDG 3) of good health and well-being, and its progress has stagnated since 2015 with an average increase in the UHC index by only 3 points (World Health Organization). As a result, there are still over 1 billion people who struggle with catastrophic health expenditures, which are healthcare payments that push households towards the poverty line. Although theoretically, healthcare is a necessity for which demand is inelastic to consumer price, healthcares services are still viewed as luxury goods in practice and inaccessible to many indigent populations. Several studies show that both developed and developing economies are experiencing rising healthcare costs due to factors such as low per Capita income, a rapidly aging population and the spread of non-communicable diseases, which aren't considered as a health hazard and therefore not as prioritized in treatment plans (Yanqiu Hou1 et al., 2025; Christian Hagist and Laurence). Kotlikoff, 2005). Three of the following pillars measure the policy's effectiveness: number of services covered, the proportion of the population covered, and the cost coverage. To assess the pillar of financial coverage of UHC, this study asks: *How do Different Financing Mechanisms of Universal Health Care Affect the Affordability in Developed vs Developing Countries?*

This study is a comparative analysis between developed and developing countries that have significantly progressed in their UHC implementation phase. Although OECD countries who have established UHC systems showed efficient spending on services, greater access, and a healthier workforce, there is still mixed evidence regarding the effect of different funding schemes on countries at all development statuses. Although developed economies have higher per Capita income, leading to greater use and accessibility towards different financing options, government budgetary strategies still vary among these nations.

To investigate this research question, I conduct cross-country regression analysis using a panel data set of 73 countries from 2000 to 2022. I construct a financial burden metric to proxy household-level affordability, and compare its effect from two major financing strategies: social health insurance contributions and tax-based funding. The model also includes foreign aid, year and regional fixed effects to fully assess domestic capacity and to account

for temporal and geographic heterogeneity in UHC implementation.

Literature Review

Many studies so far have shown that healthcare expenditures are significantly correlated with per Capita income. In a study conducted by Callander, Fox, and Lindsay (2019), findings show that the proportion of households dealing with catastrophic health expenditures in Australia is negatively associated with the income decile. Results from another case study in Brazil show that without a public sector, the rate of unmet medical needs amongst the poor would be much higher (Cataife & Courtemanche, 2011). However, income levels are not the only factor: UHC policies in developed economies target financial protection towards either the elderly or to those having non-communicable diseases (Sen, 2005). This is the main reason why the public sector has been significant in reducing the healthcare consumption on households worldwide.

Similarly to out-of-pocket expenditures, most private insurance schemes create unnecessary cost burdens to severely ill or poor people due to relatively higher premiums, which creates an adverse selection problem where people are pretending to be healthier than they are to purchase cheaper plans while not getting the coverage they deserve (Cutler, 2002 ; Grunow & Nuscheler, 2013). For example, the RAND Health Insurance Experiment showed that the largest spending drop occurred for those who purchased either the free plan or the 25% coinsurance plan, which were most likely used by the highest income group (Manning et al., 1987).

Previous literature also shows that taxes and social health insurance are the two most common approaches to reducing disparities in healthcare access. Younsi and Bechtini (2024) argue that an increased government resource mobilization- especially through taxation- leads to a proportional rise in public health expenditure. In Thailand, taxes were considered the first-best policy tool for UHC expansion, with social health insurance serving as the second-best alternative since coverage gaps re-emerge due to unemployment or retirement

(Tangcharoensathien et al., 2013). However, Gabani, Mazumdar, and Suhrcke (2022) found that countries that have transitioned into government financing systems experienced a larger decrease in out-of-pocket expenditures (OOPs) than those who have transitioned into social health insurance schemes. Existing literature also suggests that development status are strongly correlated to economic institutions that view feasibility of tax-based systems and social health insurance differently, with studies stating that social health programs are less affordable in some developing economies due to the predominant informal sector while other evidence suggests that corruption and embezzlement of government funds lead to the misallocation of tax revenues, diverting resources away from essential public health services and undermining healthcare outcomes in developing countries (Delavallade, 2006; Gupta et al., 2000; Tanzi & Davoodi, 2002). Overall, few studies provide a cross-country comparison of how the relative effectiveness of these financing mechanisms may vary by development status. This study contributes to the literature by filling this gap using a panel analysis across 73 countries.

Data/Setting

This study draws on a panel dataset from the WHO Global Health Expenditure Database, covering 194 countries between 2000 and 2022. Key indicators include Current Health Expenditure (CHE) per Capita, Out-of-Pocket Expenditure (OOP) per Capita, and Voluntary Health Insurance (VHI) as a percentage of CHE, all expressed in U.S. dollars. To complement this information, additional demographic and economic variables such as regional classification and income group were retrieved from the World Bank's World Development Indicators in order to use them controls.

Once the data had been imported into R studio, the data cleaning process involved reformatting the dataframe in a way that placed the indicators in column form and the years as value under one 'Year' column. The next step involved narrowing down the sample based on a cost burden threshold: only countries where the cost burden variable less than 50% of CHE in the years 2021 and 2022 were retained. The reason why only these two years

were considered is that I wanted to prioritize recent progression of the UHC policy in terms of financial coverage. This threshold serves as a proxy for progress toward Universal Health Coverage (UHC), aiming to focus the analysis on countries where healthcare systems are more effective at shielding households from catastrophic expenditures.

In order to compute my tax variable, 4 separate datasets on government tax revenue, government other revenue, general government health expenditure as a % of CHE and total population were obtained from the World Bank Development Indicators' data explorer. I performed the same data cleaning procedures to obtain the wide format. Then, I merged all four datasets by the country and the year column and filtered out the country variable based on the countries in my main dataset.

Model/Theory

The economic intuition behind this model is that social health insurance schemes and tax-based health expenditures, which both fall under public health insurance, reduce the financial strain on individuals. To investigate the relationship between healthcare financing mechanisms and the financial cost burden of healthcare on households, I estimate a set of ordinary least squares (OLS) regressions since I'm interested in capturing the average effect of tax-based funding and social health insurance contributions on the household cost burden in developed and developing countries, which is a continuous dependent variable. All three specifications are level-level due to the presence of missing values that were not subjected to full treatment for which the log transformation is unfeasible. Specifically, some missing values had been treated through country-mean imputation, and the remaining missing values were kept since the corresponding countries had valuable data other variables of interest. The contradicting evidence from past literature regarding the impact of financing mechanisms in developed and developing countries calls for a broader cross-country analysis that compares the effectiveness of each mechanism in lowering the per Capita financial burden between developing and developed countries. The dependent variable is Cost_Burden, which has been created taking existing indicators in the following way:

$$\text{Cost_Burden} = \frac{\text{OOP (per Capita)} + \text{VHI Payments (per Capita)}}{\text{CHE (per Capita)}}$$

$$\text{VHI Payments (per Capita)} = \frac{\text{VHI as a \% of CHE}}{100} \times \text{CHE (per Capita)}$$

This variable aims to capture the metric for the third pillar of the Universal Health Coverage policy, which is financial coverage. The main independent variables of interest are Social_Health, which represents social health insurance contributions as a % of current health expenditure and Tax_Health, representing health expenditures as a proportion of government tax revenue. The taxation variable was created as follows.

$$\text{Tax_Health} = \frac{\text{General Government Health Expenditure (GGHE)} \times \text{Tax Revenue (LCU)}}{\text{General Government Expenditure}}$$

The full derivation of the tax variable in the equation is shown in Appendix A.

There are 3 sets of specifications with 6 equations in total. The first set of equations analyze the correlation between the cost burden and social health insurance and taxation separately for developed countries by filtering the full sample to only countries for whom the developed dummy variable is equal to 1. The second set of equations is identical to the first specification, except the sub-sample contains only developing countries. These are the model specifications:

$$\begin{aligned} costburden = {} & \beta_0 + \beta_1 Social_Health + \beta_2 Ext_Health + \gamma_1 LatinAmerica \\ & + \gamma_2 NorthAmerica + \gamma_3 Europe + \gamma_4 Africa \\ & + \gamma_5 EastAsia + \gamma_6 SouthAsia + \gamma_7 SouthEAasia \\ & + \gamma_8 MiddleEast + \lambda_1 Year + \mu \end{aligned} \tag{1}$$

$$
\begin{aligned}
costburden = \beta_0 + \beta_1 Tax_Health + \beta_2 Ext_Health + \gamma_1 LatinAmerica \\
+ \gamma_2 NorthAmerica + \gamma_3 Europe + \gamma_4 Africa \\
+ \gamma_5 EastAsia + \gamma_6 SouthAsia + \gamma_7 SouthEAasia \\
+ \gamma_8 MiddleEast + \lambda_1 Year + \mu
\end{aligned} \tag{2}
$$

$$
\begin{aligned}
costburden = \beta_0 + \beta_1\ Tax_Health + \beta_2\ Social_Health + \beta_3\ Ext_Health + \beta_4\ (Social_Health \times Tax_Health) \\
+ \gamma_1\ LatinAmerica + \gamma_2\ NorthAmerica + \gamma_3\ Europe + \gamma_4\ Africa \\
+ \gamma_5\ EastAsia + \gamma_6\ SouthAsia + \gamma_7\ SouthEAAsia + \gamma_8\ MiddleEast \\
+ \lambda_1\ Year + \mu
\end{aligned} \tag{3}
$$

Where *Ext_Health* is treated as a control to account for the level of dependence on foreign aid and to capture a country's domestic capacity to fund its own healthcare system. Other control variables include regional fixed effects and year fixed effects.

Results

Empirical Findings

The results on Table 1 show that a one percentage point increase in social health insurance contributions relative to current health expenditure (CHE) leads to an approximately 1.5 percentage point increase in the cost burden relative to CHE per Capita in developed countries and for developing countries, it is a 0.17 percentage point increase. However, the positive estimates on Social_Health and External_Health do appear counterintuitive, as they are assumed to decrease the cost burden on the household. One suggestion for the positive direction is reverse causality. This means that past values of cost burdens are associated with an increased effort towards social health insurance initiatives. The possibility of inverse

causality has been tested for by running separate regressions of Social_Health and Tax_Health on lagged values of Cost_Burden and the same controls used in the main regressions (view Appendix C).

The positive estimate on CB_lag on Table 4 shows that a one-unit increase in the household cost burden from the previous year increases social health contributions by 0.178 units in developed countries and by 0.055 units in developing countries. On another note, the negative direction of all the year estimates suggest that relative to the year 2000, social health contributions have gone down in developed countries.

Table 1: Regression Results for the Model of Cost Burden Against Social Health Insurance

Regressor	**Developed**	**Developing**
Intercept	-0.0896	16.172***
	(1.273)	(0.337)
Social_Health	1.546***	0.171***
	(0.037)	(0.018)
Ext. Health	0.007	-0.301***
	(0.027)	(0.004)
N	4968	20976
R^2	0.5843	0.365

$^{***}\, p < 0.01,\ ^{**}\, p < 0.05,\ ^{*}\, p < 0.1$

Table 2 shows that a standard deviation (SD) increase in tax-based healthcare expenditure also leads to an increase in the per Capita cost burden by 1.546 standard deviations in developed countries and by 0.171 standard deviations in developing countries. The effect of taxes on the cost burden also appears counterintuitive due to its positive direction, which is why another set of regressions has been used to regress Tax_Health against CB_lag. Table 4 (view Appedix C) shows that a one SD increase in the household cost_burden from the previous year increases taxation by 0.003 standard deviations in developed countries and decreases taxation in developing countries by 0.05 standard deviations. Compared to the year-dummy estimates in Table 4, developed countries demonstrate an increased utilization in taxation over the years, whereas developing countries show a positive trend in tax-based policies until 2009, after which there is a continuous decline in tax-based funding until 2021.

Relative to 2000, there has been another significant increase in tax-based health expenditure.

Table 2: Regression Results for the Model of Cost Burden Against Taxation

Regressor	**Developed**	**Developing**
Intercept	-1.229***	-0.639***
	(0.094)	(0.023)
Tax_Health	0.017	0.090***
	(0.057)	(0.005)
Ext_Health	-0.330***	-0.393***
	(0.037)	(0.005)
N	4968	20976
R^2	0.437	0.397

$^{***}\, p < 0.01,\ ^{**}\, p < 0.05,\ ^{*}\, p < 0.1$

The main variable of interest in Table 3 is the interaction term between Tax_Health and Social_Health. Specifically, a 1 SD incease in either financing mechanism decreases the impact of the other financing mechanism on cost burden by -0.039 standard deviations in developed economies. Considering that the interaction effect is negative without considering a lagged effect, it suggests that the two mechanisms could complement each other and bring a more immediate effect on the financial burden.

On a related note, when controlling for both financing mechanisms, external aid has a protective effect on the cost burden in developing countries, as shown that a 1 percentage point increase in Ext_Health leads to a decrease in the per Capita cost burden by 0.345 standard deviations. The positive estimate for the developed country specification could be a result of inverse causality, however this hasn't been tested since Ext_Health is not the main variable of interest under this study.

Table 3: Regression Results for the Model of Cost Burden Against Both Financing Mechanisms

Regressor	**Developed**	**Developing**
Intercept	-0.608***	0.101
	(0.081)	(0.123)
Tax_Health	0.527***	0.661***
	(0.050)	(0.173)
Social_Health	0.654***	-0.265***
	(0.015)	(0.046)
Ext_Health	0.057***	-0.345***
	(0.032)	(0.034)
Tax_Health × Social_Health	-0.039***	0.068
	(0.012)	(0.153)
N	4968	20976
R^2	0.595	0.330

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

On a related note, when omitting taxes out of the specification, Ext_Health has a smaller marginal effect on Cost_Burden in developed countries. For developing countries, a 1-unit increase in Ext_Health aid leads to a 0.301 unit decrease in cost burden. Similarly, the positive estimate for developed countries could also be related to reverse causality, but this hasn't been tested for since this is not a variable of interest.

Discussion

Although the estimates of Social_Health and Tax_Health are statistically significant at the 1% level, the findings from the specification of Social_Health and Tax_Health on CB_lag hold more economic meaning. Table 4 suggests that developed countries have a greater social health insurance contribution capacity compared to developing countries due to the higher magnitude of the estimate. In practice, developing nations are known to have a larger informal sector, where most citizens are self-employed and therefore are not part of a payroll system. Although tax-based funding increases with higher cost burden ratio

in developed countries, it has an opposite effect among developing countries. This result supports the idea that the tax-based systems in the developing countries under this sample might be less responsive to cost burden, and there might be a tradeoff between the two financing schemes. The interaction term's estimate on the pooled model also suggests that taxes and social health insurance contributions have a greater protective effect on household financial burden in developed countries, however it is associated with higher cost burden in developing countries. Without empirically testing for reverse causality, one could suggest that pooling mechanisms may take longer to reduce cost burden in developing economies due to administrative barriers and the need to establish regulatory frameworks. In contexts where both tax-based and SHI systems are underdeveloped or poorly coordinated, combining them may result in fragmentation, inefficiencies, and overlapping coverage gaps- leading to a higher cost burden. In such cases, a mixed approach may perform worse than relying on one well-implemented system. To improve this study, incorporating more microeconomic control variables related to the elderly population rate, the urbanization rate, institutional quality could have further isolated the pure effect of the financing mechanisms on the financial burden. Although this study does not interact financing variables with time, it is possible that the effects of tax-based and SHI mechanisms on cost burden as systems mature. An improvement to this study may involve interacting the financing variables with the year fixed effects to assess how each financing method varies over time.

Conclusion

This study set out to evaluate how different financing mechanisms of Universal Health Coverage (UHC) impact household-level affordability across both developed and developing countries. Using panel data from 73 countries between 2000 and 2022, the cross-country regression analysis compared the effects of tax-based funding and social health insurance contributions on a constructed financial burden metric, while controlling for foreign aid, time, and regional fixed effects. The findings reveal that there is an overall greater average effect of social health insurance contributions on household cost burden in developed countries and a larger average effect of taxation on cost burden in developing countries. Furthermore, the

lagged cost burden models reveal that cost burden has a larger positive impact on social health initiatives in developed economies while higher cost burden is associate with a relatively larger and negative effect on taxation initiatives, which suggests that social health initiatives are more responsive to higher financial burden among the countries in the sample. Although this analysis provides valuable insights, it is not without limitations. Country-level economic and institutional factors- key elements in UHC implementation- were not directly included in the model. Future research could explore these dimensions further, as well as delve into micro-level datasets for a more granular understanding of household experiences under different UHC models.

Use of Generative AI and AI-assisted tools

During the preparation of my thesis, I used Chat GPT and Perplexity AI for assistance on the R code for my empirical analysis and the LaTex code for data visualization. After using this tool/service, I reviewed and edited the content as needed and take full responsibility for the content of my thesis.

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Appendices

Appendix A: Derivation of the Tax_Health variable (R Code)

```
#Step 1: Creating a dataset for total government revenue

    TaxRev <- API_GC_TAX_TOTL_CN_DS2_en_excel_v2_1253 %>% pivot_longer(cols=
    starts_with("20"), names_to= "Year", values_to= "Value")
TaxRev

TaxRev <- TaxRev %>% pivot_wider(names_from= `Indicator Name`, values_from= Value
    )
TaxRev #Tax revenue data set

OtherRev <- API_GC_REV_GOTR_CN_DS2_en_excel_v2_1183 %>%
  filter(`Country Name` %in% countries_under_50_both_years)

OtherRev <- API_GC_REV_GOTR_CN_DS2_en_excel_v2_1183 %>% pivot_longer(cols= starts
    _with("20"), names_to= "Year", values_to="Value")
OtherRev

OtherRev <- OtherRev %>% pivot_wider(names_from= `Indicator Name`, values_from=
    Value)
OtherRev #Other government revenue data set

TaxRev$Year <- as.numeric(as.character(TaxRev$Year))
OtherRev$Year <- as.numeric(as.character(OtherRev$Year))

#Merging the two datasets to get total government revenue

TotalRev <- merge(TaxRev, OtherRev,
                     by = c("Country Name", "Country Code", "Year"))

TotalRev <- TotalRev %>%
  mutate(across(-c(`Country Name`, `Country Code`), as.numeric))

TotalRev[] <- lapply(TotalRev, function(x) {
  if (is.numeric(x)) {
    x[is.na(x)] <- mean(x, na.rm = TRUE)
  }
  return(x)
})

#Step 2: Creating a general government expenditure variable
```

```
TotalRev$GGE <- TotalRev$`Tax revenue (current LCU)`+ TotalRev$`Grants and other
    revenue (current LCU)`#Here I assume that total government revenue= total
    government expenditure

gghe_percapita <- API_SH_XPD_GHED_PP_CD_DS2_en_excel_v2_17513 %>% pivot_longer(
    cols= starts_with("20"), names_to="Year", values_to="Value")
gghe_percapita

gghepercapita_final <- gghe_percapita %>% pivot_wider(names_from= `Indicator Name
    `, values_from= Value)
gghepercapita_final #Aggregate health expenditure as a % of GGE dataset

totalpop <- API_SP_POP_TOTL_DS2_en_excel_v2_76243 %>% pivot_longer(cols= starts_
    with("20"), names_to="Year", values_to="Value")
totalpop

totalpop_final <- totalpop %>% pivot_wider(names_from= `Indicator Name`, values_
    from= Value)
totalpop_final #Total population dataset

#Merging health expenditure and population datasets

gghepercapita_final$`Country Name` <- as.character(gghepercapita_final$`Country
    Name`)
gghepercapita_final$`Country Code` <- as.character(gghepercapita_final$`Country
    Code`)
gghepercapita_final$Year <- as.character(gghepercapita_final$Year)

totalpop_final$`Country Name` <- as.character(totalpop_final$`Country Name`)
totalpop_final$`Country Code` <- as.character(totalpop_final$`Country Code`)
totalpop_final$Year <- as.character(totalpop_final$Year)

TotalExp <- merge(gghepercapita_final, totalpop_final,by = c("Country Name", "
    Country Code", "Year"))
TotalExp

#Step 3: Creating a total government health expenditure variable

TotalExp$TotalGGHE <- TotalExp$`Domestic general government health expenditure
    per capita, PPP (current international $)`* TotalExp$`Population, total`

TotalExp <- TotalExp %>%
  mutate(across(-c(`Country Name`,`Country Code`, Year), as.numeric))

TotalExp[] <- lapply(TotalExp, function(x) {
  if (is.numeric(x)) {
    x[is.na(x)] <- mean(x, na.rm = TRUE)
```

```
  }
  return(x)
})

#Step 4: Creating the taxation variable

TotalExp$taxhealth <- (TotalExp$AggGGHE*TotalRev$`Tax revenue (current LCU)`)/
    TotalRev$GGE

view(TotalExp)

TotalExp <- TotalExp %>%
  filter(`Country Name` %in% countries_under_50_both_years)

TotalExp <- TotalExp %>%
  rename(
    Countries= `Country Name`
  )

taxdata<- TotalExp %>%
  select(Countries, Year, taxhealth)

taxdata <- taxdata %>%
  mutate(Year= as.numeric(Year))

Merged_data <- Merged_data %>%
  left_join(taxdata, by=c("Countries", "Year"))

view(Merged_data)
```

Appendix B: Distribution of SHI and Tax Variables

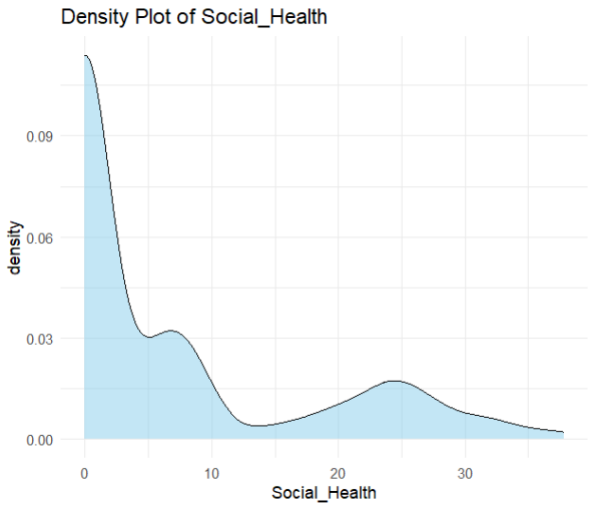

Figure 1: Screenshot 1

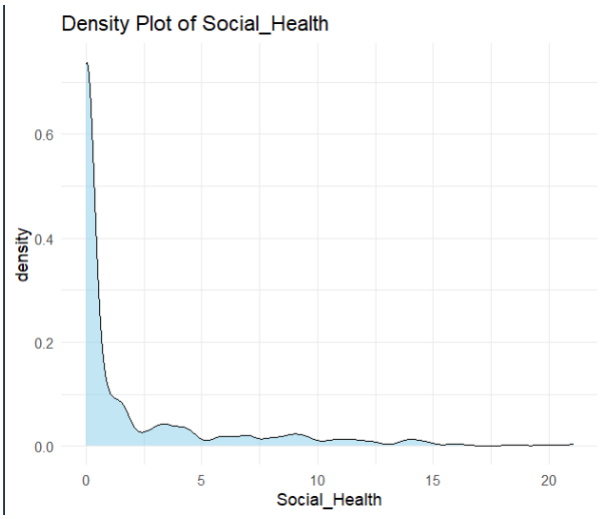

Figure 2: Screenshot 2

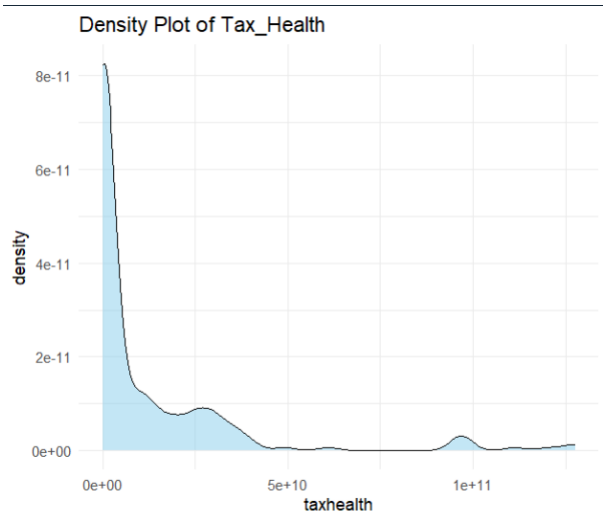

Figure 3: Screenshot 1

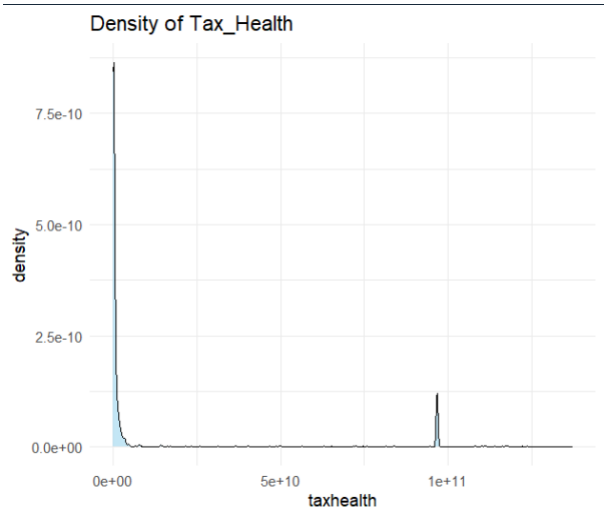

Figure 4: Screenshot 2

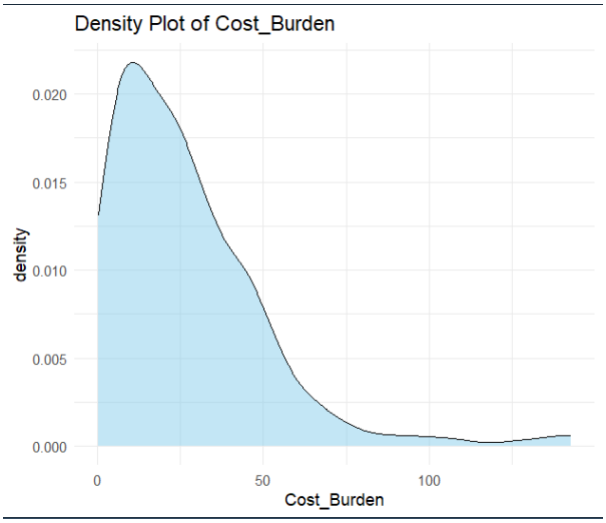

Figure 5: Screenshot 1

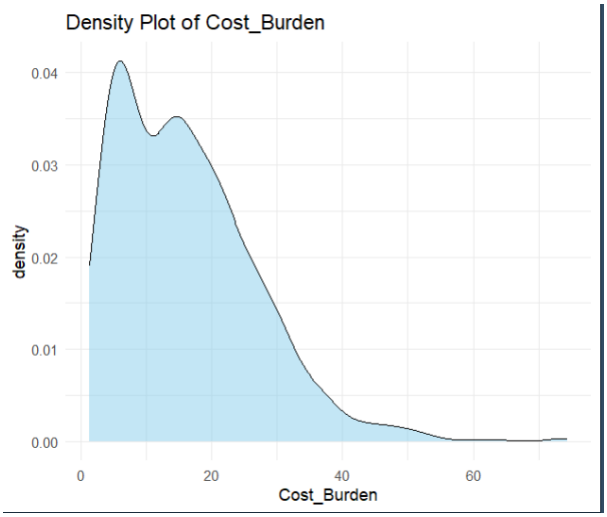

Figure 6: Screenshot 2

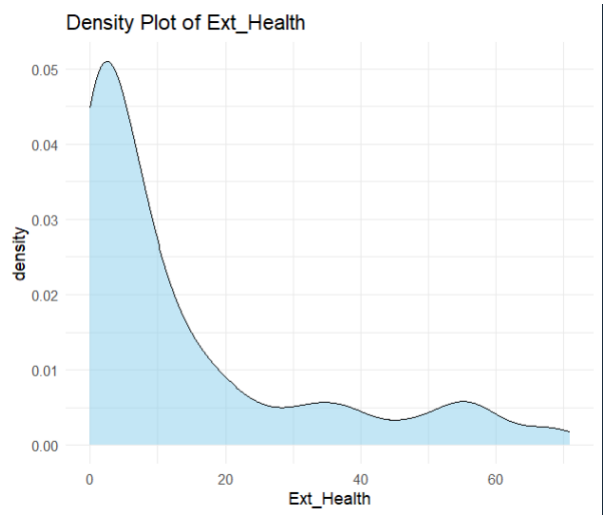

Figure 7: Screenshot 1

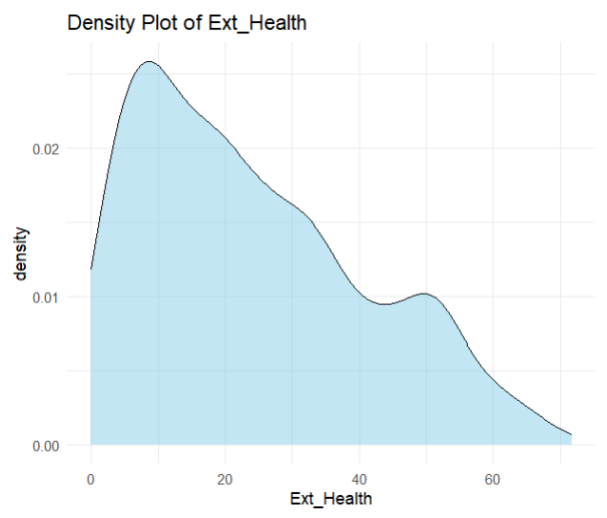

Figure 8: Screenshot 2

Appendix C: Table of the Lagged Cost Burden Models

	Developed		Developing	
	Social_Health	Tax_Health	Social_Health	Tax_Health
Intercept	0.043	-0.561***	0.418**	0.037
	(0.389)	(0.023)	(0.136)	(0.031)
CB_lag	0.178***	0.003	0.055***	-0.048***
	(0.004)	(0.004)	(0.002)	(0.008)
*Year*2001	0.218	0.003	-0.043	0.001**
	(0.508)	(0.029)	(0.159)	(0.037)
*Year*2006	-1.699***	0.076**	-0.281	0.025
	(0.509)	(0.029)	(0.159)	(0.037)
*Year*2011	-2.729***	0.147***	-0.490**	-0.025
	(0.512)	(0.029)	(0.1600)	(0.037)
*Year*2016	-0.610**	0.270***	-0.561***	0.006
	(0.508)	(0.029)	(0.160)	(0.038)
*Year*2021	-1.353**	0.464***	-0.600**	-0.058***
	(0.509)	(0.029)	(0.160)	(0.038)
*Year*2022	-1.282*	3.635***	-0.626**	3.282***
	(0.509)	(0.029)	(0.165)	(0.038)
N	4959		20938	
R^2	0.584	0.437	0.365	0.38

Note: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$

Appendix D: General Relationship of Cost Burden and the Two Financing Mechanisms

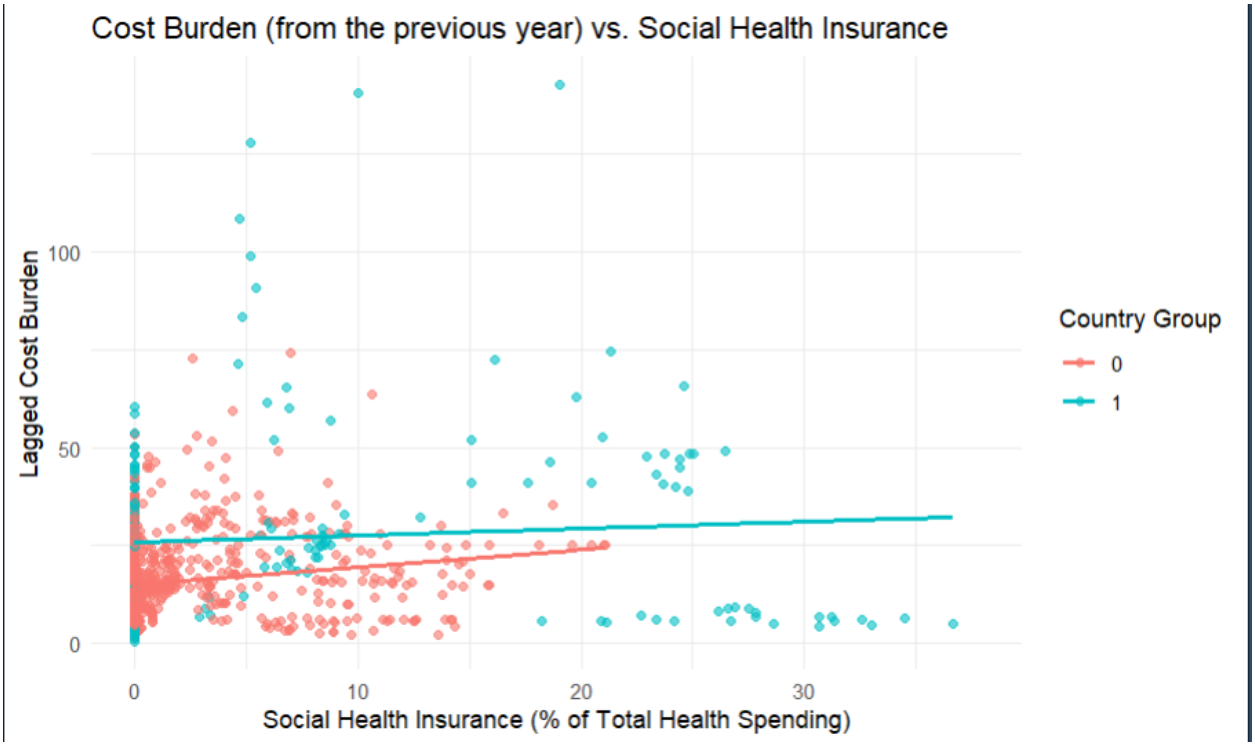

Figure 9: Scatter plot showing the relationship between SHI as a % of total health expenditure and the cost burden as a % of total health expenditure per Capita

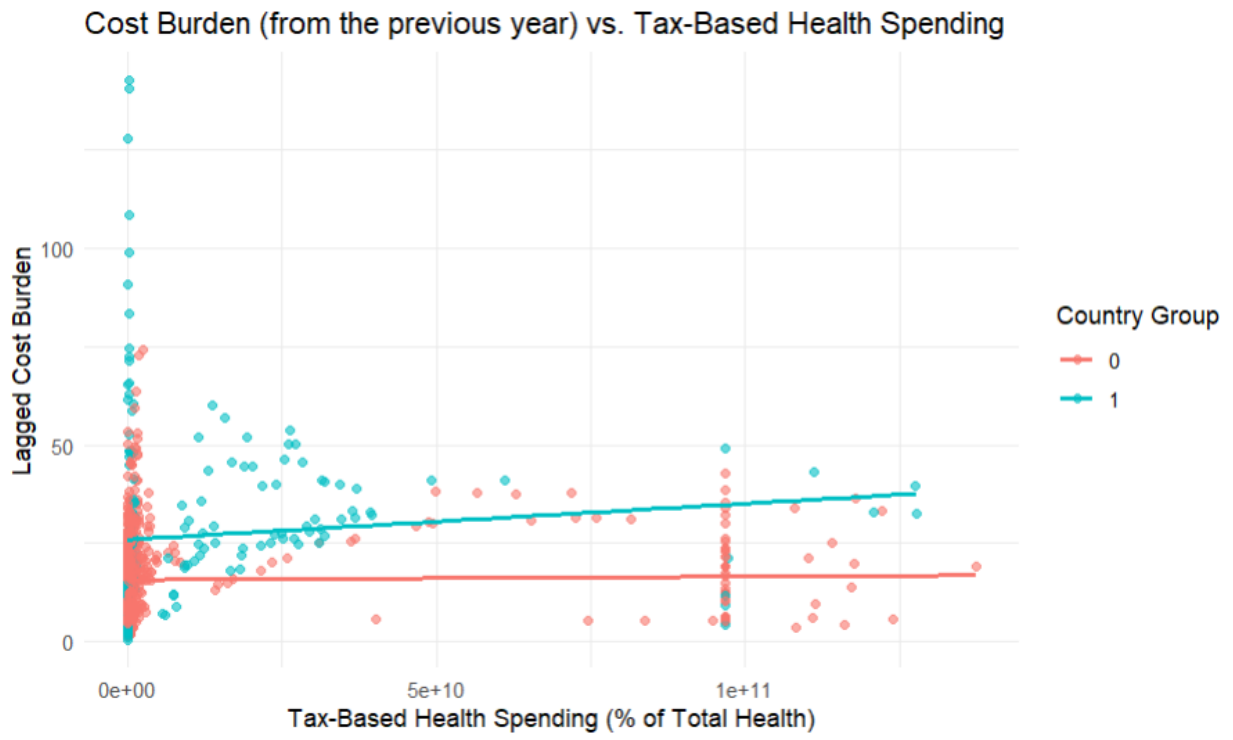

Figure 10: Scatter plot showing the relationship between tax-based health expenditure and the cost burden, both standardized to have a mean of 0 and a standard deviation of 1